



Computer Networks

IT351

Project-Phase 2

Deadline: Day 09/08/2026 @ 23:59

[Total Mark is 15]

Name: Layan Alghamdi

ID: S240014014

Name: Alanoud Alfalah

ID: S240061825

Name: Rawan Almalki

ID: S240051772

Name: Safanah Alfahaid

ID: S240031066

Instructions:

- You must submit two separate copies (**one Word file and one PDF file**) using the Assignment Template on Blackboard via the allocated folder. These files **must not be in compressed format**.
- It is your responsibility to check and make sure that you have uploaded both the correct files.
- Zero mark will be given if you try to bypass the SafeAssign (e.g., misspell words, remove spaces between words, hide characters, use different character sets, **convert text into image** or languages other than English or any kind of manipulation).
- Email submission will not be accepted.
- You are advised to make your work clear and well-presented. This includes filling your information on the cover page.
- You must use this template, failing which will result in zero mark.
- You **MUST** show all your work, and text must not be converted into an image, unless specified otherwise by the question.
- Late submission will result in ZERO mark.
- The work should be your own, copying from students or other resources will result in ZERO mark.
- Use **Times New Roman** font for all your answers.

Project Instruction

Project Objective:

Design, configure, and test a network setup in **Packet Tracer** to successfully transmit data packets between devices in different LANs across two separate WANs (WAN 1 and WAN 2).

Group Details

- Group Size = Maximum of 3 Members.
- One group member (group leader/coordinator) must **submit the project report** on blackboard. Marks will be given based on your submission and quality of the contents.

Project Report

- The project is divided into two phases:
 - Phase 1 (Mid project Progress Report)
 - Phase 2 (Final project Report)
- This report is for Phase 2. In this report, you should complete all the requirements for Phase 2.
- Each Report will be evaluated according to the marking criteria

Learning Outcome(s):

Demonstrate protocol configuration, network-addressing schemes and analyze packet transmission for the Local and Wide Area Networks.

Project Description (Common to Phase 1 and Phase 2)

A university operates two campuses: a Main Campus and a Branch Campus. Each campus network is treated as a separate WAN site. Your task is to design and configure, in Cisco Packet Tracer, a network that successfully transmits data packets from a device in any LAN of WAN 1 (Main Campus) to a device in any LAN of WAN 2 (Branch Campus). Each WAN contains 2 LANs, and each LAN has 3 end devices.

Network Design Requirements:

1. WAN 1 (Main Campus):

- Contains LAN 1 (Computer Lab) and LAN 2 (Admin Office).
- Each LAN has 3 devices (PCs or laptops) connected through a switch.
- Assign private IP addresses to the devices in each LAN (e.g., use 172.16.10.0/24 for LAN 1 and 172.16.20.0/24 for LAN 2).
- Connect each LAN to a router (Router 1).

2. WAN 2 (Branch Campus):

- Contains LAN 3 (Library) and LAN 4 (Faculty Office).
- Each LAN has 3 devices (PCs or laptops) connected through a switch.
- Assign private IP addresses to the devices in each LAN (e.g., use 172.16.30.0/24 for LAN 3 and 172.16.40.0/24 for LAN 4).
- Connect each LAN to a router (Router 2).

3. WAN Connection:

- Connect Router 1 and Router 2 using a serial link to simulate a WAN connection.
- Assign IP addresses to the serial interfaces (e.g., 10.10.10.1/30 for Router 1 and 10.10.10.2/30 for Router 2).

Configuration Details:

1. Assign IP Addresses:

- Configure IP addresses, subnet masks, and default gateways for all devices in each LAN. Example:
 - LAN 1: 172.16.10.0/24, Default Gateway: 172.16.10.1 (Router 1's interface).

- LAN 2: 172.16.20.0/24, Default Gateway: 172.16.20.1 (Router 1's interface).
- LAN 3: 172.16.30.0/24, Default Gateway: 172.16.30.1 (Router 2's interface).
- LAN 4: 172.16.40.0/24, Default Gateway: 172.16.40.1 (Router 2's interface).

2. Router Configuration:

- Configure IP addresses on the LAN and WAN interfaces of both routers.

Example:

- Router 1:
 - LAN 1 Interface: 172.16.10.1/24
 - LAN 2 Interface: 172.16.20.1/24
 - WAN Interface (Serial): 10.10.10.1/30
- Router 2:
 - LAN 3 Interface: 172.16.30.1/24
 - LAN 4 Interface: 172.16.40.1/24
 - WAN Interface (Serial): 10.10.10.2/30

3. Enable Routing:

- Configure static routes or a dynamic routing protocol (e.g., RIP) on both routers to ensure communication between WAN 1 and WAN 2.
-

Phase 2 Requirements

Testing and Verification

1. Use the ping command to test connectivity between a device in any LAN of WAN 1 and a device in any LAN of WAN 2.
2. Verify that data packets are successfully transmitted (0% packet loss).

Questions to Answer (10 Marks — 2 marks each)

Answer all five questions. For each question, you must include (a) a clear screenshot from your Packet Tracer project and (b) the commands used at that instance (e.g., IOS CLI commands or the configuration steps applied). Screenshots without the corresponding commands/steps will receive ZERO marks for that question.

1. **Creating nodes:** Show a screenshot of the complete topology with all nodes created (12 end devices, 4 switches, 2 routers). List the device types selected and any steps/commands used to name the devices (e.g., hostname configuration). (2 marks)
2. **Giving connections in the LAN:** Show a screenshot of the LAN connections between end devices, switches, and routers. State the cable types used for each connection and why. (2 marks)
3. **Assigning IP addresses:** Show screenshots of IP address, subnet mask, and default gateway assignment on at least one end device in each LAN, and the commands used to configure the router LAN interfaces (e.g., interface, ip address, no shutdown). (2 marks)
4. **Creating the WAN connection:** Show a screenshot of the serial link between Router 1 and Router 2, and the commands used to configure the serial interfaces and enable routing (static routes or a dynamic routing protocol such as RIP), including clock rate if applicable. (2 marks)
5. **Successful data transmission using the ping command:** Show screenshots of successful ping tests between a device in WAN 1 and a device in WAN 2 (both directions), with the exact ping commands used and the results (replies received, 0% loss). (2 marks)

Written Reflection — End of Project Report (5 Marks)

The project report must include a Reflection section (300–500 words) as the last section before the Project Artifacts section. This section must be written in the students' own words and should address some of the following:

1. Describe one specific challenge or obstacle you encountered during this project and explain in detail how you identified and resolved it.
2. Explain why you chose your specific approach/method (e.g., static routing vs. RIP, your IP addressing scheme) over at least one alternative you considered, and what tradeoffs informed that decision.
3. If you used any AI tools during this project (e.g., ChatGPT, Claude, Copilot), disclose which tool(s), for what specific purpose (e.g., debugging, grammar checking, brainstorming), and how you verified or modified the output.
4. What would you do differently if you were to redo this project with more time or resources?

Version History / Commit Log Requirements (Common to Both Reports)

Students must document and submit their project history/timestamps using a live, editable link:

- **Written report:** Submit a shared Google Doc or Word Online link (sharing set to "Anyone with the link can view") along with your final PDF report. The link must show the progress of the students' work with timestamps/screenshots.
- **Code/data/configuration files:** Submit Packet Tracer files (.pkt) and any configuration scripts via a GitHub or GitLab repository link (the coordinator may choose a repository suitable for the course). The repository must show a commit history reflecting incremental progress. Commit regularly (at minimum once per week), with clear, descriptive commit messages.

A separate section in the report must be added and named "Project Artifacts" to include all links, for example:

- Live report document (with edit history): [link]
- GitHub/GitLab repository (with commit history): [link]
- Packet Tracer project file (.pkt): [link or attached]

What NOT to do: Students must not write their report elsewhere and paste the final version in at the last minute. They also must not make a single commit containing the entire project. Both are treated as red flags regardless of the quality of the final work.

Important Notes

- **AI use:** Students may use AI tools (ChatGPT, Claude, Copilot, Gemini, etc.) only for brainstorming approaches, debugging small errors, or proofreading text. AI must not be used to generate analysis, write configuration logic, or produce substantial portions of the report or project files. Any permitted AI use must be disclosed in the Reflection section. Undisclosed AI use, or AI use beyond this scope (disclosed or not), will result in a zero grade for the entire project.
- Any submission lacking a valid, inspectable version/commit history, or showing evidence of AI generation or plagiarism, will receive a zero grade for the entire project.
- Failure to submit the mid progress report as required, or submission of reports found to be fabricated, copied, or AI-generated, will result in a zero grade for the entire project.
- The final report must be submitted as a PDF, together with the live document link, the repository link, and the Packet Tracer (.pkt) file.
- The edit history of the live document and the commit history of the repository must show incremental progress across both phases (Phase 1 and Phase 2).

Heading خطأ! استخدم علامة التبويب "الصفحة الرئيسية" لتطبيق
على النص الذي ترغب في أن يظهر هنا 1

Pg. 8

Network Topology

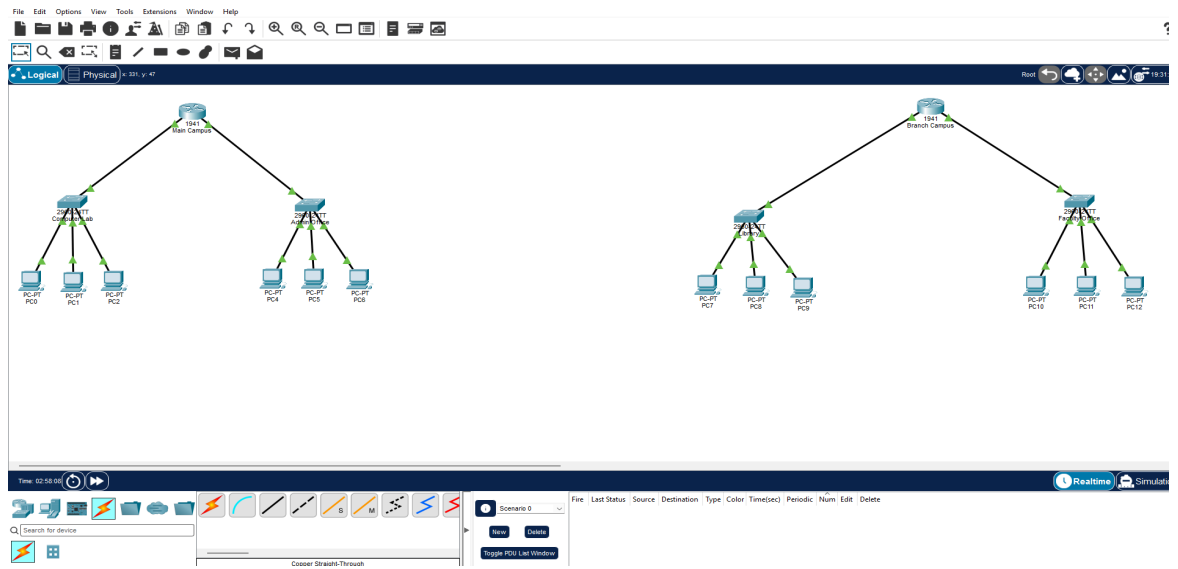


Figure 1. Initial Network Topology

Devices Used in the Network

Device Type	Model	Quantity
Router	Cisco 1941	2
Switch	Cisco 2960-24TT	4
PC	PC-PT	12

Commands Used to Change Device Names

Router (**Main Campus**):

```
enable
configure terminal
hostname MainCampus
end
copy running-config startup-config
```

Router (**Branch Campus**):

```
enable
configure terminal
hostname Branch Campus
end
copy running-config startup-config
```

Switch (**Computer Lab**) :

```
enable
configure terminal
hostname ComputerLab
end
copy running-config startup-config
```

Switch (**Admin Office**):

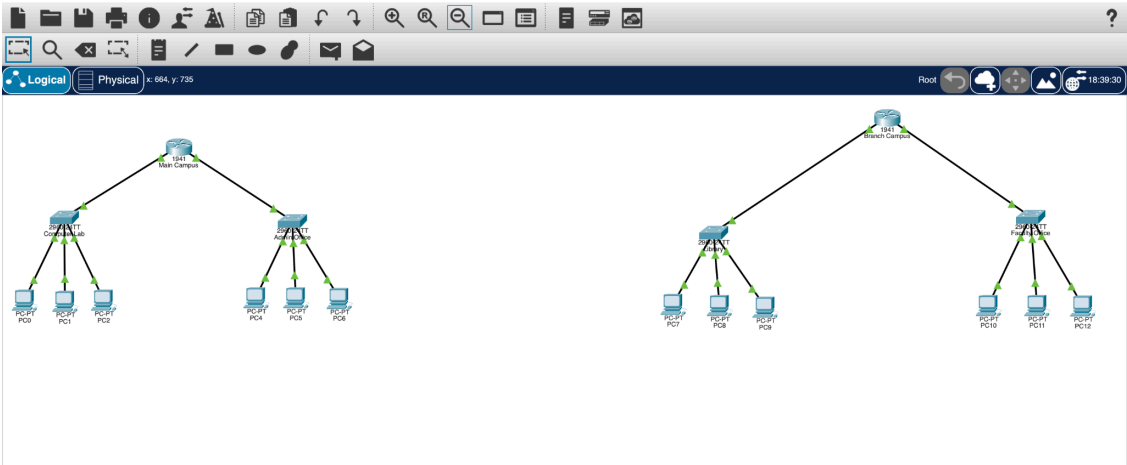
```
enable
configure terminal
hostname AdminOffice
end
copy running-config startup-config
```

Switch (**Library**):

```
enable
configure terminal
hostname Library
end
copy running-config startup-config
```

Switch (**Faculty Office**):

```
enable
configure terminal
hostname FacultyOffice
end
copy running-config startup-config
```



Connection	Cable Type	Reason
PC to Switch	Copper Straight-Through	Used to connect different types of devices (PC and Switch).
Switch to Router	Copper Straight-Through	Used to connect different types of devices (Switch and Router).
LAN	Network	Subnet Mask Default Gateway

Heading خطأ! استخدم علامة التبويب "الصفحة الرئيسية" لتطبيق
على النص الذي ترغب في أن يظهر هنا 1

ComputerLab	172.16.10.0/24	255.255.255.0	172.16.10.1
AdminOffice	172.16.20.0/24	255.255.255.0	172.16.20.1
Library	172.16.30.0/24	255.255.255.0	172.16.30.1
FacultyOffice	172.16.40.0/24	255.255.255.0	172.16.40.1

Heading خطأ! استخدم علامة التبويب "الصفحة الرئيسية" لتطبيق على النص الذي ترغب في أن يظهر هنا 1

The image displays two screenshots of a network configuration interface, likely from a Packet Tracer simulation. Both screenshots show the 'IP Configuration' tab for a specific PC.

PC4 Configuration:

- Interface: FastEthernet0
- IP Configuration:
 - ☐ DHCP
 - ☒ Static
 - IPv4 Address: 172.16.20.2
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 172.16.20.1
 - DNS Server: 0.0.0.0
- IPv6 Configuration:
 - ☐ Automatic
 - ☒ Static
 - IPv6 Address: (empty)
 - Link Local Address: FE80::207:ECFF:FE4E:6822
 - Default Gateway: (empty)
 - DNS Server: (empty)
- 802.1X:
 - ☐ Use 802.1X Security
 - Authentication: MDS
 - Username: (empty)
 - Password: (empty)

PC0 Configuration:

- Interface: FastEthernet0
- IP Configuration:
 - ☐ DHCP
 - ☒ Static
 - IPv4 Address: 172.16.10.3
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 172.16.10.1
 - DNS Server: 0.0.0.0
- IPv6 Configuration:
 - ☐ Automatic
 - ☒ Static
 - IPv6 Address: (empty)
 - Link Local Address: FE80::201:43FF:FE14:4030
 - Default Gateway: (empty)
 - DNS Server: (empty)
- 802.1X:
 - ☐ Use 802.1X Security
 - Authentication: MDS
 - Username: (empty)
 - Password: (empty)

Heading خطأ! استخدم علامة التبويب "الصفحة الرئيسية" لتطبيق
على النص الذي ترغب في أن يظهر هنا 1

PC7

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 172.16.30.2

Subnet Mask 255.255.255.0

Default Gateway 172.16.30.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::20C:FFFF:FE8B:A64

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

PC10

Physical Config **Desktop** Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 172.16.40.2

Subnet Mask 255.255.255.0

Default Gateway 172.16.40.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::2E0:80FF:FE9C:298A

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

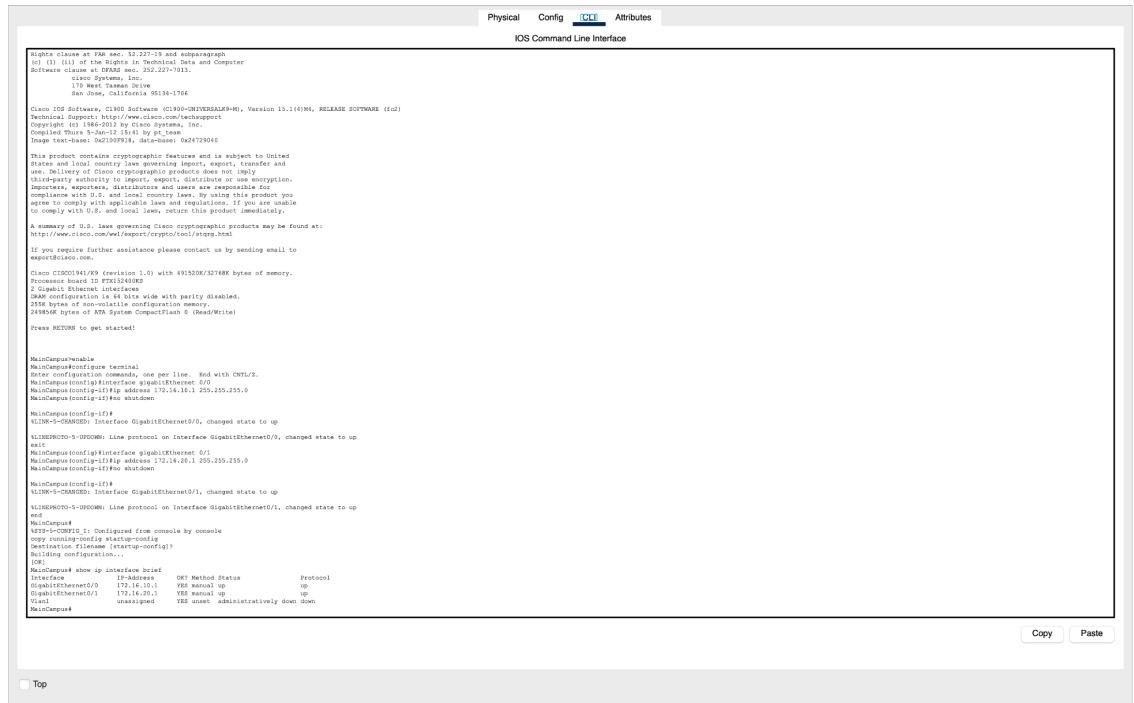
Authentication MD5

Username

Password

Heading خطأ! استخدم علامة التبويب "الصفحة الرئيسية" لتطبيق على النص الذي ترغب في أن يظهر هنا 1

Pg. 15



```
Physical Config CLI Attributes
IOS Command Line Interface

rights clause at FAR sec. 52.227-13 and subparagraph
(c) (1) (1) of the Rights in Technical Data and Computer
Software clause at FAR sec. 52.227-7013.
Cisco Systems, Inc.
170 West Tasman Drive
San Jose, California 95134-1706

Cisco IOS Software, C1900 Software (C1900-UNIVERSALK9-M), Version 15.1(4)M6, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2015 by Cisco Systems, Inc.
Compiled Thu 4-Jun-15 15:41 by pt_tam
Image text-base: 0x00000000, data-base: 0x00000000

This product contains cryptographic features and is subject to United
States and local country law governing import, export, transfer and
use. Delivery of Cisco cryptographic products does not imply
illegitimate authority to export, reexport, distribute or use encryption.
Importers, exporters, distributors and users are responsible for
compliance with U.S. and local country law. By using this product you
agree to comply with applicable laws and regulations. If you are unable
to comply with U.S. and local law, return this product immediately.

A summary of U.S. laws governing Cisco cryptographic products may be found at:
http://www.cisco.com/wai/exportcrypto/tool/etapp.html
If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco C1900-100 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX1240802
0 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
259K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

MainConfig#enable
MainConfig#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
MainConfig(config)#interface gigabitEthernet 0/0
MainConfig(config-if)#ip address 172.16.10.1 255.255.255.0
MainConfig(config-if)#no shutdown

MainConfig(config-if)#
%LINK-0-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
MainConfig(config-if)#ip address 172.16.20.1 255.255.255.0
MainConfig(config-if)#no shutdown

MainConfig(config-if)#
%LINK-0-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
MainConfig#
VTP>>>CONFIG 1: Configured from console by console
copy running-config startup-config
Write memory (startup-config)?
Building configuration...
[OK]
MainConfig#show ip interface brief
Interface IP-Address OK Method Status Protocol
GigabitEthernet0/0 172.16.10.1 VRF manual up up
GigabitEthernet0/1 172.16.20.1 VRF manual up up
Vlan1 unassigned VRF unset administratively down down
MainConfig#
```

enable
configure terminal
interface gigabitEthernet 0/0
ip address 172.16.10.1 255.255.255.0
no shutdown
exit
interface gigabitEthernet 0/1
ip address 172.16.20.1 255.255.255.0
no shutdown
end
copy running-config startup-config

Heading خطأ! استخدم علامة التبويب "الصفحة الرئيسية" لتطبيق على النص الذي ترغب في أن يظهر هنا 1

Pg. 16

```
IOS Command Line Interface

Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 172.16.30.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 172.16.40.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
Router#
Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
GigabitEthernet0/0 172.16.30.1 VLS manual up
GigabitEthernet0/1 172.16.40.1 VLS manual up
Serial0/0/0 unassigned VLS unset administratively down down
Serial0/0/1 unassigned VLS unset administratively down down
Vlan1 unassigned VLS unset administratively down down
Router#

Router# end is now available

Press RETURN to get started.
```

enable
configure terminal
interface gigabitEthernet 0/0
ip address 172.16.30.1 255.255.255.0
no shutdown
exit
interface gigabitEthernet 0/1
ip address 172.16.40.1 255.255.255.0
no shutdown
end
copy running-config startup-config

Heading خطأ! استخدم علامة التبويب "الصفحة الرئيسية" لتطبيق على النص الذي ترغب في أن يظهر هنا 1

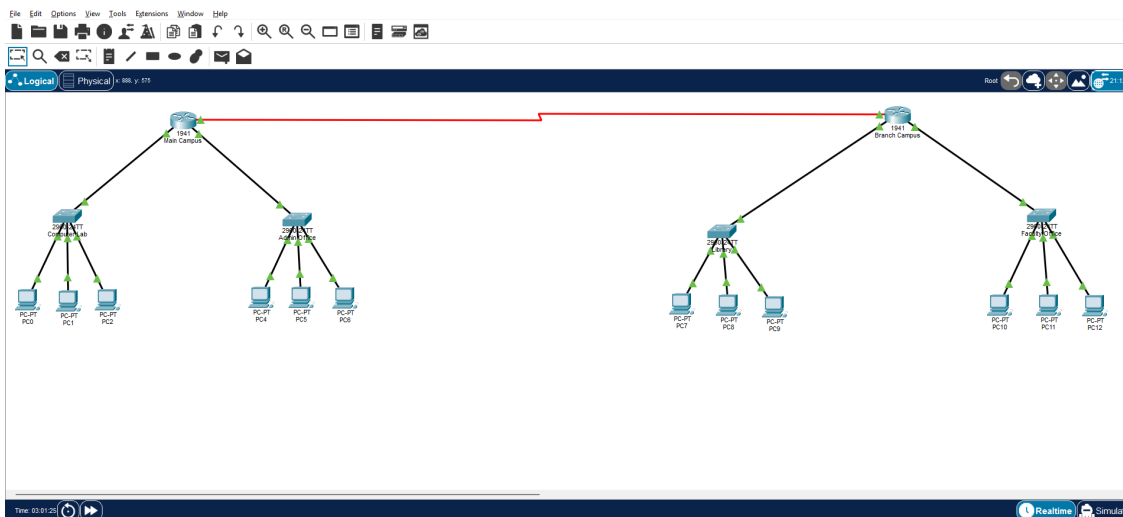


Figure - Final Design of the Network

Commands for configuring the serial interface and enabling routing:

```

Main Campus
Physical Config CLI Attributes
Cisco IOS Command Line Interface

MainCampus> enable
MainCampus> configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
MainCampus(config)# interface Serial0/1/0
MainCampus(config-if)# ip address 10.10.1.255 255.255.255.252
MainCampus(config-if)# clock rate 56000
MainCampus(config-if)# no shutdown
MainCampus(config-if)# exit
MainCampus(config)#
%LINK-3-UPDOWN: Interface Serial0/1/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up

MainCampus>
MainCampus> enable
MainCampus> configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
MainCampus(config)# ip route 172.16.0.0 255.255.255.0 10.10.1.2
MainCampus(config)# ip route 172.16.40.0 255.255.255.0 10.10.1.2
MainCampus(config)#
MainCampus>
RTB-S-CORP01_1_ Configured from console by console
  
```

Router 1 (**Main Campus**):

/Serial interface configuration

enable

configure terminal

interface serial0/1/0

ip address 10.10.10.1 255.255.255.252

clock rate 64000

no shutdown

exit

/Static routes

ip route 172.16.30.0 255.255.255.0 10.10.10.2

ip route 172.16.40.0 255.255.255.0 10.10.10.2

Heading خطأ! استخدم علامة التبويب "الصفحة الرئيسية" لتطبيق على النص الذي ترغب في أن يظهر هنا 1

```

Branch Campus
Physical Config CLI Attributes
IOS Command Line Interface

1 GigabitEthernet interfaces
2 200-based serial ports/async network interfaces
3 ROM configuration is 4k bits wide with parity disabled.
4 256K bytes of non-volatile configuration memory.
5 2495168 bytes of ATA System CompactFlash 0 (Ssd/Flash)
6
7 Press RETURN to get started!

VLINEP0070-4-UDONM: Line protocol on Interface GigabitEthernet0/0, changed state to up
VLINEP0070-4-UDONM: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router#config hostname BranchCampus
BranchCampus(config)# interface serial10/1/0
BranchCampus(config-if)# ip address 10.10.10.2 255.255.255.252
BranchCampus(config-if)# no shutdown
BranchCampus(config-if)#
VLINEP-1-CHANGED: Interface Serial10/1/0, changed state to up
VLINEP0070-4-UDONM: Line protocol on Interface Serial10/1/0, changed state to up

BranchCampus con0 is now available

Press RETURN to get started.

BranchCampus# enable
BranchCampus# configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
BranchCampus# configure ip route 172.16.10.0 255.255.255.0 10.10.10.1
BranchCampus# configure ip route 172.16.10.0 255.255.255.0 10.10.10.1
BranchCampus# configure ip
BranchCampus#
8079-0-COPYING_1: Configured from console by console

```

Router 2 (**Branch Campus**):

/Serial interface configuration

enable

configure terminal

interface serial10/1/0

ip address 10.10.10.2 255.255.255.252

no shutdown

exit

/Static routes

ip route 172.16.10.0 255.255.255.0 10.10.10.1

Heading خطأ! استخدم علامة التبويب "الصفحة الرئيسية" لتطبيق على النص الذي ترغب في أن يظهر هنا 1

```
ip route 172.16.20.0 255.255.255.0 10.10.10.1
```

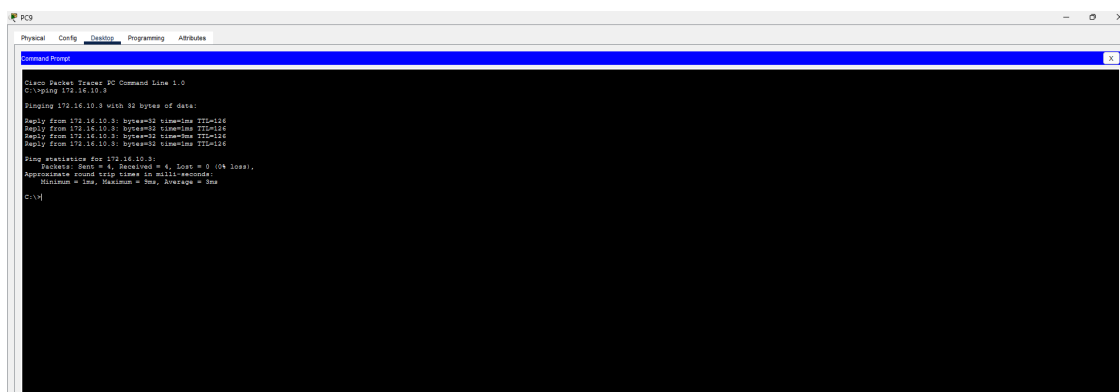
Data Transmission using the Ping Command:



```
PC4
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ping 172.16.30.2
Pinging 172.16.30.2 with 32 bytes of data:
Reply from 172.16.30.2: bytes=32 time=1ms TTL=126
Reply from 172.16.30.2: bytes=32 time=1ms TTL=126
Reply from 172.16.30.2: bytes=32 time=1ms TTL=126
Reply from 172.16.30.2: bytes=32 time=1ms TTL=126
Ping statistics for 172.16.30.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms
C:\>
```

Figure - From Main Campus to Branch Campus

```
ping 172.16.30.2
```



```
PC9
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ping 172.16.10.3
Pinging 172.16.10.3 with 32 bytes of data:
Reply from 172.16.10.3: bytes=32 time=1ms TTL=126
Reply from 172.16.10.3: bytes=32 time=1ms TTL=126
Reply from 172.16.10.3: bytes=32 time=1ms TTL=126
Reply from 172.16.10.3: bytes=32 time=1ms TTL=126
Ping statistics for 172.16.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms
C:\>
```

Figure - From Branch Campus to Main Campus

ping **Version History / Commit Log**

The following internal record summarizes the main stages represented in the submitted report and Packet Tracer file.

Version	Phase	Work Completed	Status
1.0	Phase 1	Created the initial topology, selected devices, and named routers and switches.	Completed
1.1	Phase 1	Connected LAN devices and prepared the IPv4 addressing plan for all four LANs.	Completed
2.0	Phase 2	Configured LAN interfaces, serial addressing, clocking, and static routes.	Completed
2.1	Phase 2	Verified bidirectional communication with successful ping tests and 0% packet loss.	Completed
2.2	Final	Reviewed commands, added reflection and artifacts, and prepared Word and PDF files.	Completed

172.16.10.3

Reflection

Finishing this project shifted how we think about troubleshooting, mostly because the whole layout only made sense if every piece behaved like part of a single system. The biggest hurdle showed up after we had all four LANs in place: getting consistent, dependable communication between the Main Campus and the Branch Campus. Inside each LAN, things worked without much effort. Crossing the WAN was different. Those first end-to-end tests forced us to slow down and verify the basics carefully, router interfaces, the serial connection, default gateways, and the routing table entries that would actually carry traffic from one side to the other.

To track it down, we went back to the addressing table, pulled interface status and configuration outputs from the routers, and traced the route one step at a time. Both ends of the serial link were brought up properly, the DCE side was set with a 64000 clock rate, and we double-checked every static route that needed to exist. After that, we reran the pings in both directions. Seeing four replies and 0% packet loss was the clear sign that the LAN settings and the WAN link were finally working together the way they were supposed to.

We chose static routing instead of RIP because the network is small, just two routers and four known LANs, and nothing about the design called for dynamic updates. With static routes, the forwarding path stayed obvious: each destination network and its next hop could be confirmed directly, without guessing what the protocol had learned. RIP might help if the topology expanded later since it cuts down on manual route edits, but here it would have meant extra configuration and constant routing updates that didn't add much value. For addressing, we kept the LANs on /24 subnets because they were simple to manage and easy to tell apart. On the serial link, we used a /30 since a point-to-point connection only needs two usable addresses. The tradeoff is straightforward: static routing doesn't scale well, and adding more campuses or networks would mean more manual changes.

Any AI help stayed limited to polishing the final text, checking formatting, and spotting minor presentation issues in the way commands were written. It wasn't used to run Packet Tracer or to do the actual configuration work for the group. Whenever a technical change came up during editing, we only kept it after matching it against the real topology, router output, IP settings, and, most importantly, the successful ping results.

If we were to do the project again, we'd test and document every stage immediately after finishing it instead of waiting until the end. That would mean cleaner screenshots for interface status, routing tables, and ping output, plus a more structured troubleshooting checklist. We'd also keep more

frequent local copies of the report and the Packet Tracer file. With that approach, errors would be easier to isolate, and the final submission would end up stronger and more consistent.

Project Artifacts

Cisco Packet Tracer project file (.pkt): IT351_FinalDesign.pkt (attached separately).

Final written report: IT351_Project-Phase_2_Final.docx.

Final PDF report: IT351_Project-Phase_2_Final.pdf.

Google Doc: [Project-Phase2](#)

GitHub/GitLab repository: [GitHub](#)

Packet Tracer project file (.pkt): [Packet Tracer](#)

Final Submission

Submission Item	Status
Final Word report (.docx)	Prepared
Final PDF report (.pdf)	Prepared
Cisco Packet Tracer project (.pkt)	Attached separately